# ERIC FITHIAN

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## EDUCATION

**University of Colorado Boulder** — Boulder, CO
B.S. in Computer Science, *summa cum laude*, GPA: 4.0/4.0 — Aug. 2022 – May 2025
Dean's List all three years; graduated one year early

## RESEARCH EXPERIENCE

**University of Chicago** — Chicago, IL
Center for Applied Artificial Intelligence
*Research Professional / Predoctoral Fellow* — July 2025 – Present

- **DRY-SFT** (Prof. X.Y. Han): Designed a reward-free self-supervised fine-tuning method that trains on sequentially generated diverse solutions, increasing output diversity and improving high pass@$k$ performance by more than 10 percentage points across HumanEval+, MBPP+, and DS-1000.
- **Reward horizons** (Prof. Suproteem K. Sarkar): Investigating how short-horizon evaluation misses substantial later corrective feedback in software engineering, and whether critic models trained on delayed code corrections can better align post-training rewards with long-term objectives.
- **Mode collapse** (Prof. X.Y. Han): Contributed to the idea and method, implementation, experiments, and analysis in a study of how supervised fine-tuning calibrates LLM output diversity.
- **AI-assisted operations research** (Prof. Rad Niazadeh): Coauthored a study of LLM pipelines for extracting open problems, reviewing subsequent literature, and generating and verifying candidate solutions.
- **Medical AI**: Coauthored a study of vision-language models for surgical tool detection with Prof. X.Y. Han; research on chest X-ray foundation models with Prof. Sarah Jabbour.
- **Earlier collaborations**: Built NLP pipelines covering 11 million news documents to study journalist ideology with Prof. Jacob Conway; applied pretrained embeddings to consumer choice models with Prof. Giovanni Compiani.

**University of Colorado Boulder, Human Biobehavior Signals Lab** — Boulder, CO
*Research Assistant to Prof. Theodora Chaspari* — June 2024 – Oct. 2025

- First-authored research on multimodal hirability prediction combining transformer-based NLP, OpenFace, FaceNet, and multimodal fusion; presented a poster at ACM ICMI 2025 in Canberra, Australia.

## PUBLICATIONS

**Eric Fithian**, Theodora Chaspari. *Leveraging Pre-Trained Transformers and Facial Embeddings for Multimodal Hirability Prediction in Job Interviews.* Proceedings of the 27th ACM International Conference on Multimodal Interaction (ICMI), 2025. [Paper]

## WORKSHOP PAPERS

**Eric Fithian**, Suproteem K. Sarkar. *The Reward Horizon Problem: Learning From Delayed Feedback.* 2026. Accepted poster, TAE (Trust-AI-Eval): Can We Trust AI Evaluation?, NeurIPS 2026 workshop (non-archival). Under review at ICLR 2027. [Workshop] [Manuscript]

**Eric Fithian**, Kirill Skobelev, X.Y. Han. *Don't Repeat Yourself: Self-Supervised Fine-Tuning for Coverage.* 2026. Accepted to the Transitioning from Pre-Training to Post-Training workshop, NeurIPS 2026 (non-archival). Under review at ICLR 2027. [arXiv]

Kirill Skobelev, **Eric Fithian**, X.Y. Han. *Fine-Tuning Fixes Mode Collapse and Over-Dispersion in LLMs.* 2026. Accepted to the Transitioning from Pre-Training to Post-Training workshop, NeurIPS 2026 (non-archival). Under review at ICLR 2027. [arXiv]

**Eric Fithian**, Rad Niazadeh, Pranav Nuti. *Solving Open Problems in Operations Research Using AI*. 2026. [Paper] [Project]
Poster at Can AI do Theory?, STOC TheoryFest 2026; presented at AI-Driven Research in Econ-CS, EC 2026 (non-archival workshops).

Kirill Skobelev, **Eric Fithian**, Yegor Baranovski, Jack Cook, Sandeep Angara, Shauna Otto, Zhuang-Fang Yi, John Zhu, Neeraj Mainkar, Margaux Masson-Forsythe, Daniel A. Donoho, X.Y. Han. *A Comparative Study in Surgical AI: Potential and Limitations of Data, Compute, and Scaling*. 2026. Accepted to the GenAI for Health workshop, NeurIPS 2026 (non-archival). Under review at *NEJM AI*. [arXiv]

## WORKING PAPERS AND PREPRINTS

**Eric Fithian**, Kirill Skobelev. *DELM: a Python toolkit for Data Extraction with Language Models*. arXiv preprint, 2025. [arXiv]

## PROJECTS AND SOFTWARE

**DELM**: Co-developed an open-source Python toolkit for reproducible, scalable LLM-based structured data extraction, including validation, batch processing, caching, and cost tracking. [GitHub] [PyPI] [Docs]

**VizWiz VQA Challenge**: Designed a ViT+BERT architecture with bidirectional cross-attention for visual question answering; first place among approximately 150 graduate students as an undergraduate. [GitHub]

## HONORS AND AWARDS

- First Place, VizWiz VQA Challenge 2025
- Computer Science departmental nominee, CU Boulder College of Engineering Silver Medal (one senior nominated per department annually) 2025
- CU Boulder Chancellor's Recognition Award (straight A's; fewer than 50 recipients) 2025
- *summa cum laude* 2025
- Dean's List, all three years 2022–2025

## INDUSTRY EXPERIENCE

**Spectrum / Charter Communications** Denver, CO
*Software Engineering Intern* June 2024 – Aug. 2024

- Developed a mobile data usage simulation service; reverse-engineered legacy Perl and Shell scripts to fix critical production bugs.

## SERVICE AND TEACHING

Volunteer Software Engineer, Blueprint Boulder 2023 – 2024
Private Tutor, Computer Science and Calculus 2022 – 2024

## SKILLS

**Programming**: Python (PyTorch, Transformers, pandas, NumPy), Java, C++, SQL, Git, LaTeX.
**Research**: LLM post-training, NLP, multimodal AI, computer vision, reinforcement learning.
**Systems**: High-performance computing (HPC), distributed multi-GPU training.